Formative Assessment 6

8.21 A population consists of the four numbers 3, 7, 11, and 15. Consider all possible samples of size 2 that can be drawn with replacement from this population. Find (a) the population mean, (b) the population standard deviation, (c) the mean of the sampling distribution of means, and (d) the standard deviation of the sampling distribution of means. Verify parts (c) and (d) directly from (a) and (b) by using suitable formulas.

8.34 Find the probability that of the next 200 children born, (a) less than 40% will be boys, (b) between 43% and 57% will be girls, and (c) more than 54% will be boys. Assume equal probabilities for the births of boys and girls.

8.49 The credit hour distribution at Metropolitan Technological College is as follows:

x	6	9	12	15	18
$p(x)$	0.1	0.2	0.4	0.2	0.1

Find μ and σ^2. Give the 25 (with replacement) possible samples of size 2, their means, and their probabilities.

Solve:

FAQ - Statistical Theory

8.21

a) $\mu = \frac{\Sigma x}{n}$

$= \frac{3+7+11+15}{4}$

$= \frac{36}{4} = 4$

* "n" number of score

* "N" number of sample size

b) $\sigma^2 = \frac{\Sigma (x-\mu)^2}{n}$

$= \frac{(3-9)^2 + (7-9)^2 + (11-9)^2 + (15-9)^2}{4}$

$= \frac{80}{4} = 20$

$\sigma = \sqrt{\sigma^2} = \sqrt{20} \approx 4.47$

c) $\mu_{\bar{x}} = 9$ (fact based from the book $\mu_{\bar{x}} = \mu$)

d) $\sigma^2_{\bar{x}} = \frac{\sigma^2}{N} = \frac{20}{2} = 10$

8.34

given: $n = 200$, $p = 0.5$, $q = 1 - p = 0.5$ (since assumed the probabilities of boys and girls are equal, $p = 0.5$)

solve: $\mu_p = p = 0.5$

$\sigma_p = \sqrt{\frac{pq}{n}} = \sqrt{\frac{(0.5)(0.5)}{200}} \approx 0.0353$

a) $z = \frac{\hat{p} - \mu_p}{\sigma_p}$

$= \frac{0.40 - 0.50}{0.0353}$

≈ -2.83 (use z table for looking A of -2.83)

$A = 0.00233$ 0.23% chance of less than 40% will be boys

b) $z_1 = \frac{0.43 - 0.50}{0.0353} \approx -1.98$ $A = 0.02385$

$z_2 = \frac{0.57 - 0.50}{0.0353} \approx 1.98$ $A = 0.97615$

$0.97615 - 0.02385 = 0.95227$

95.52% chance of between 43% & 57% will be girls

c) $z = \frac{0.54 - 0.50}{0.0353} \approx 1.13$

$A = 0.27076$

$= 1 - 0.27076$ (since we looking for more than 54%)

$= 0.12924$ 12.92% chance that 54% will be boys

8.49 $\mu = \sum x \cdot P(x)$

$= (6 \times 0.1) + (9 \times 0.2) + (12 \times 0.4) + (15 \times 0.2) + (18 \times 0.1)$

$= 12$

$\sigma^2 = \sum x^2 \cdot P(x) - \mu^2$

$= (36 \times 0.1) + (81 \times 0.2) + (144 \times 0.4) + (225 \times 0.2) + (324 \times 0.1)$

$= 154.8 - 12^2$

$= 10.8$

The 25 (with replacement) possible samples and their means

Sample	**Mean ($\bar{x}$)**	**Sample**	**Mean ($\bar{x}$)**	**Sample**	**Mean ($\bar{x}$)**	**Sample**	**Mean ($\bar{x}$)**	**Sample**	**Mean ($\bar{x}$)**
(6,6)	6	(9,6)	7.5	(12,6)	9	(15,6)	10.5	(18,6)	12
(6,9)	7.5	(9,9)	9	(12,9)	10.5	(15,9)	12	(18,9)	13.5
(6,12)	9	(9,12)	10.5	(12,12)	12	(15,12)	13.5	(18,12)	15
(6,15)	10.5	(9,15)	12	(12,15)	13.5	(15,15)	15	(18,15)	16.5
(6,18)	12	(9,18)	13.5	(12,18)	15	(15,18)	16.5	(18,18)	18

Probabilities of Sample Means

$\bar{x}$	$P(\bar{x})$	**Probability**
6	0.1×0.1	0.01
7.5	$2(0.1 \times 0.2)$	0.04
9	$2(0.1 \times 0.4) + (0.2 \times 0.2)$	0.12
10.5	$2(0.1 \times 0.4) + 2(0.2 \times 0.4)$	0.20
12	$2(0.1 \times 0.1) + 2(0.2 \times 0.2) + (0.4 \times 0.4)$	0.26
13.5	$2(0.2 \times 0.4) + 2(0.1 \times 0.2)$	0.20
15	$2(0.1 \times 0.4) + (0.2 \times 0.2)$	0.12
16.5	$2(0.1 \times 0.2)$	0.04
18	0.1×0.1	0.01
Sum		**1**